



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

AMAZON NEPTUNE PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Databases

TOTAL: 10 QUESTIONS

Q1 What type of database is Amazon Neptune and what are its core strengths?

Threat Model

Q6 How do you monitor and tune slow queries in Amazon Neptune?

Observability

Q2 How does Amazon Neptune guarantee consistency and handle transactions?

Architecture

Q7 What backup and restore strategies does Amazon Neptune support?

Lifecycle

Q3 What index types does Amazon Neptune support and when do you use each?

Configuration

Q8 How do you handle schema migrations in Amazon Neptune?

Compliance

Q4 How do you design a schema or data model in Amazon Neptune?

Hardening

Q9 What security features does Amazon Neptune provide?

Testing

Q5 How does Amazon Neptune scale horizontally?

SSO / IdP

Q10 What are common anti-patterns when using Amazon Neptune in production?

Tuning



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Amazon Neptune is a relational, document,

Mitigation

Amazon Neptune is a relational, document, key-value, graph, or time-series store. Its strengths such as ACID guarantees, horizontal scale, or flexible schema guide the workloads it is best suited for.

A6 Amazon Neptune provides a slow query

Observability

Amazon Neptune provides a slow query log, explain/explain plan, or profiling tools to surface inefficient queries. Common fixes include adding indexes, rewriting queries, or increasing cache size.

A2 Amazon Neptune provides either full ACID

Primitives

Amazon Neptune provides either full ACID transactions or eventual consistency depending on its CAP theorem positioning. Transaction support varies from none (simple K/V stores) to serializable isolation (relational DBs).

A7 Amazon Neptune supports logical backups

Automation

Amazon Neptune supports logical backups (export SQL or BSON), physical backups (filesystem snapshot), and continuous replication to a standby. Point-in-time recovery requires WAL/oplog archiving on top of backups.

A3 Amazon Neptune offers B-tree, hash, full-text,

Hardening

Amazon Neptune offers B-tree, hash, full-text, spatial, or composite indexes depending on the engine. Choose based on query patterns: B-tree for range queries, hash for equality lookups, full-text for search.

A8 Schema migrations in Amazon Neptune are

Governance

Schema migrations in Amazon Neptune are managed with versioned migration files applied in order by a migration tool. Migrations should be idempotent and tested in staging before production to avoid downtime.

A4 Schema design in Amazon Neptune involves

Gotchas

Schema design in Amazon Neptune involves normalizing relations (relational DB) or denormalizing for access patterns (document/key-value). Understanding query needs upfront prevents costly migrations later.

A9 Amazon Neptune supports role-based access

Assessment

Amazon Neptune supports role-based access control, encrypted connections (TLS), encryption at rest, and audit logging. Always restrict user privileges to the minimum needed and disable anonymous access in production.

A5 Amazon Neptune scales via replication (read

Federation

Amazon Neptune scales via replication (read replicas) for read-heavy workloads and sharding (partitioning data by key) for write-heavy ones. Some Amazon Neptune implementations handle this automatically; others require manual configuration.

A10 Common mistakes include selecting all

Optimization

Common mistakes include selecting all columns instead of needed fields, missing indexes on frequently queried columns, running migrations without backups, and storing large blobs directly in the database instead of object storage.